Quantification: 876_z76c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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| measure | value |
|---|---|
| lacuna count | 9 |
| interior lacuna count | 8 |
| frame-edge lacuna count | 1 |

Interior lacunae only: mean, median and sample SD (n = 8)
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| measure | kind | mean | median | sd | units |
|---|---|---|---|---|---|
| area_px2 | lacuna shape | 1619.29 | 1847 | 625.922 | px^2 |
| major_axis_length_px | lacuna shape | 74.8552 | 85.0968 | 19.0124 | px |
| minor_axis_length_px | lacuna shape | 28.1986 | 28.6646 | 5.4419 | px |
| aspect_ratio | lacuna shape | 2.6446 | 2.6707 | 0.4723 | unitless |
| eccentricity | lacuna shape | 0.918 | 0.9273 | 0.0333 | unitless |
| solidity | lacuna shape | 0.8303 | 0.8641 | 0.0791 | unitless |
| roots_count | headline | 8.125 | 7.5 | 3.8336 | count |
| ring_length_r30_px | headline | 533.125 | 524.5 | 160.355 | px |
| ring_length_r60_px | reported | 1506.75 | 1525.5 | 415.841 | px |
| owned_length_px | ownership-dependent | 5522.12 | 6032.4 | 2700.91 | px |
| edge_count | ownership-dependent | 405.75 | 417.5 | 204.51 | count |
| mean_edge_length_px | ownership-dependent | 13.7285 | 13.7817 | 0.7571 | px |
| perimeter_px | normalised | 230.364 | 236.066 | 41.9093 | px |
| ring_area_r30_px2 | normalised | 8291.25 | 8605 | 1296.25 | px^2 |
| ring_area_r60_px2 | normalised | 21764.9 | 22785 | 3029.55 | px^2 |
| in_frame_fraction_r30 | normalised | 0.9993 | 1 | 0.0019 | unitless |
| in_frame_fraction_r60 | normalised | 0.9836 | 1 | 0.0463 | unitless |
| ring_density_r30 | normalised | 0.064611 | 0.0729034 | 0.0174071 | px^-1 |
| ring_density_r60 | normalised | 0.0691473 | 0.0744062 | 0.0172031 | px^-1 |
| roots_per_100px_perimeter | normalised | 3.4978 | 3.5282 | 1.3127 | per 100 px |
| ring_attached_length_r30_px | network v2 | 479 | 483 | 190.349 | px |
| ring_attached_length_r60_px | network v2 | 1264.25 | 1254.5 | 563.705 | px |
| ring_length_w_r30_px | network v2 | 599.081 | 587.446 | 180.173 | px |
| ring_length_w_r60_px | network v2 | 1689.89 | 1707.53 | 469.571 | px |
| sholl_crossings_r10 | network v2 | 12.875 | 13.5 | 4.0156 | count |
| sholl_crossings_r20 | network v2 | 16.375 | 18.5 | 4.3074 | count |
| sholl_crossings_r30 | network v2 | 19.125 | 19 | 6.1047 | count |
| loop_count | loops, report only | 54.625 | 55 | 33.0062 | count |
| loop_edge_share | loops, report only | 0.4667 | 0.5121 | 0.2014 | share |
| ring_width_mean_r30_px | width | 6.0474 | 6.0889 | 0.8975 | px |

Per field
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| measure | value | units |
|---|---|---|
| analysed_area_px2 | 1.03268e+06 | px^2 |
| total_skeleton_length_px | 66370 | px |
| canalicular_length_density_per_px | 0.0642695 | px^-1 |
| junction_count | 2241 | count |
| junction_density_per_px2 | 0.00217008 | px^-2 |
| lacuna_count | 9 | count |

| | | |
|---|---|---|
| lacunae_per_px2 | 8.72e-06 | px^-2 |
| skeleton_component_count | 658 | count |
| mean_component_length_px | 100.866 | px |
| median_component_length_px | 13 | px |
| field_length_density_w_per_px | 0.0720361 | px^-1 |
| field_density_without_flagged_per_px | 0.0658093 | px^-1 |
| field_density_in_roi_per_px | | px^-1 |
| unattached_length_px | 156 | px |
| unattached_component_count | 9 | count |
| canalicular_length_density_attached_per_px | 0.0642695 | px^-1 |
| field_length_density_with_unattached_per_px | 0.0644205 | px^-1 |
| loop_count | 548 | count |
| loop_edge_share | 0.4213 | share |
| edge_count | 5001 | count |

Canalicular width, the new measure

| measure | value | units |
|---|---|---|
| width_mean_px | 6.1439 | px |
| width_median_px | 6 | px |
| width_p10_px | 2.8284 | px |
| width_p90_px | 10 | px |
| width_pixels_used | 54352 | count |

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

| lacuna_id | area_px2 | major_axis_length_px | minor_axis_length_px | aspect_ratio | eccentricity | solidity | orientation_rad | centroid_row_px | centroid_col_px | on_border |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 3529 | 196.715 | 34.7861 | 5.655 | 0.9842 | 0.5697 | -1.4629 | 16.8161 | 858.163 | True |
| 2 | 1841 | 89.0492 | 27.9031 | 3.1914 | 0.9496 | 0.8121 | 1.5494 | 146.123 | 381.449 | False |
| 3 | 2155 | 85.0968 | 33.783 | 2.5189 | 0.9178 | 0.8641 | -1.5153 | 296.21 | 913.135 | False |
| 4 | 2003 | 88.5498 | 30.0304 | 2.9487 | 0.9407 | 0.8649 | 1.5644 | 297.345 | 325.757 | False |
| 5 | 1105 | 63.6459 | 23.8313 | 2.6707 | 0.9273 | 0.806 | -1.5304 | 527.835 | 198.727 | False |
| 6 | 1847 | 70.5663 | 34.2892 | 2.058 | 0.874 | 0.8966 | -1.5649 | 538.637 | 444.666 | False |
| 7 | 422 | 38.2927 | 18.8887 | 2.0273 | 0.8699 | 0.6709 | -1.0621 | 646.408 | 104.687 | False |
| 8 | 1962 | 88.7858 | 28.6646 | 3.0974 | 0.9464 | 0.8975 | -1.4695 | 687.621 | 605.365 | False |
| 9 | | | | | | | | 925.23 | 965.577 | False |

Network per lacuna

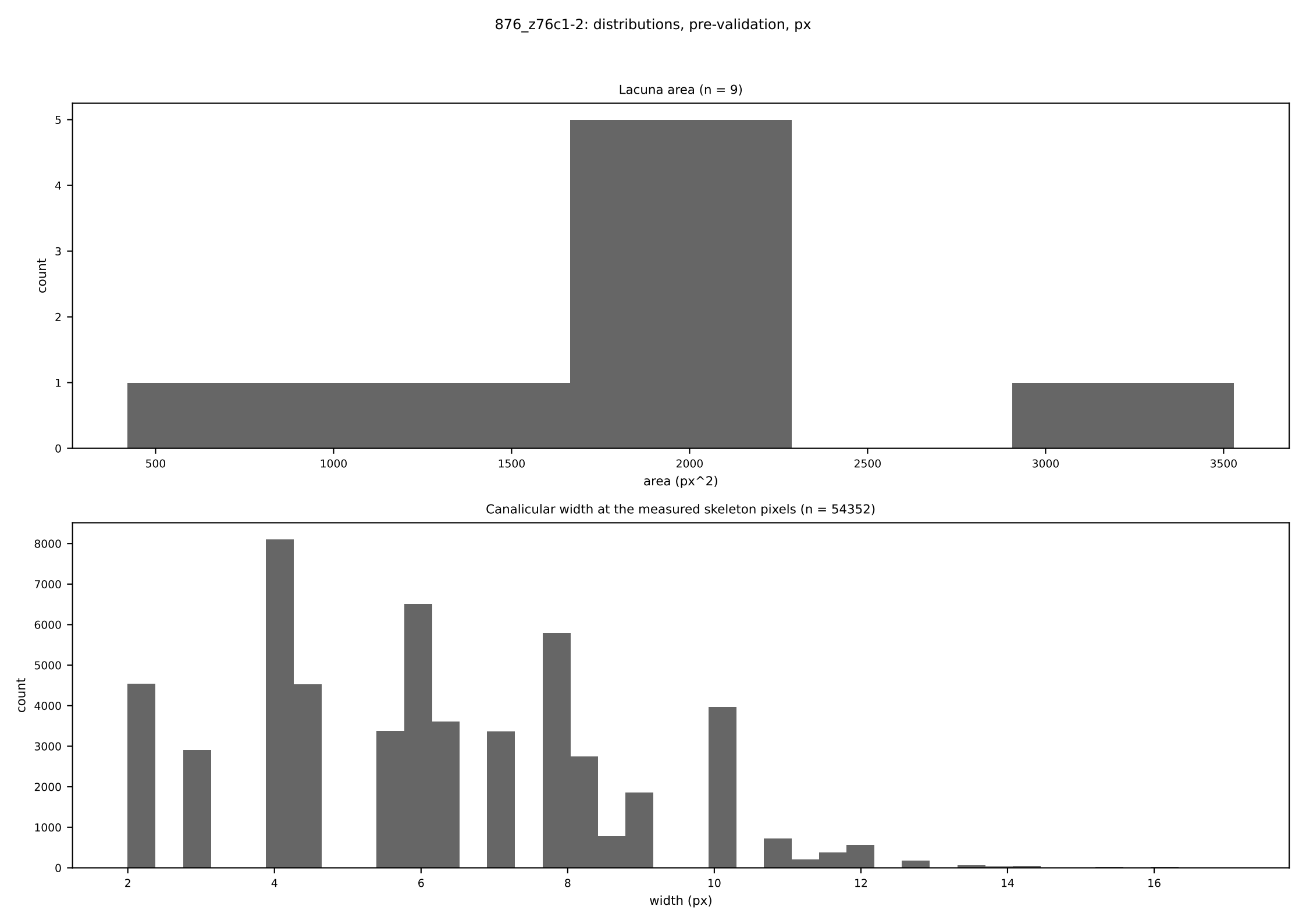
| lacuna_id | roots_count | ring_length_r30_px | ring_length_r60_px | owned_length_px | edge_count | mean_edge_length_px | ring_width_mean_r30_px |
|---|---|---|---|---|---|---|---|
| 1 | 16 | 648 | 1549 | 5435.72 | 355 | 15.3119 | 3.688 |
| 2 | 16 | 666 | 1742 | 6421.94 | 424 | 15.1461 | |
| 3 | 9 | 447 | 1298 | 6789.25 | 485 | 13.9985 | |
| 4 | 10 | 766 | 1958 | 9578.03 | 724 | 13.2293 | 6.9134 |
| 5 | 6 | 602 | 1885 | 7131.32 | 564 | 12.6442 | 6.4157 |
| 6 | 9 | 645 | 1833 | 5642.86 | 411 | 13.7296 | 6.9445 |
| 7 | 6 | 435 | 1309 | 4190.91 | 319 | 13.1376 | 5.7621 |
| 8 | 5 | 416 | 1274 | 3928.82 | 284 | 13.8339 | 5.6496 |
| 9 | 4 | 288 | 755 | 493.818 | 35 | 14.1091 | 4.5992 |

Normalised per lacuna

| lacuna_id | perimeter_px | ring_area_r30_px2 | ring_area_r60_px2 | in_frame_fraction_r30 | in_frame_fraction_r60 | ring_density_r30 | ring_density_r60 | roots_per_100px_perimeter |
|---|---|---|---|---|---|---|---|---|
| 1 | 756.927 | 11162 | 23083 | 0.6571 | 0.6039 | 0.0580541 | 0.0671057 | 2.1138 |
| 2 | 269.948 | 9365 | 24218 | 1 | 1 | 0.0711159 | 0.07193 | 5.9271 |
| 3 | 274.883 | 9487 | 24500 | 1 | 1 | 0.0471171 | 0.0529796 | 3.2741 |
| 4 | 264.392 | 9452 | 24487 | 1 | 1 | 0.0810411 | 0.0799608 | 3.7823 |
| 5 | 195.072 | 7582 | 20596 | 1 | 1 | 0.0793986 | 0.0915226 | 3.0758 |
| 6 | 210.593 | 8239 | 22047 | 1 | 1 | 0.0782862 | 0.0831406 | 4.2736 |
| 7 | 155.894 | 5824 | 17026 | 1 | 1 | 0.0746909 | 0.0768824 | 3.8488 |
| 8 | 249.12 | 8971 | 23523 | 1 | 1 | 0.0463716 | 0.0541598 | 2.0071 |
| 9 | 223.013 | 7410 | 17722 | 0.9946 | 0.8691 | 0.0388664 | 0.0426024 | 1.7936 |

Network v2 per lacuna

| lacuna_id | ring_attached_length_r30_px | ring_attached_length_r60_px | ring_length_w_r30_px | ring_length_w_r60_px | sholl_crossings_r10 | sholl_crossings_r20 | sholl_crossings_r30 |
|---|---|---|---|---|---|---|---|
| 1 | 582 | 952 | 722.63 | 1720.24 | 17 | 16 | 19 |
| 2 | 637 | 1402 | 741.529 | 1933.41 | 18 | 21 | 21 |
| 3 | 402 | 915 | 495.99 | 1435.16 | 14 | 12 | 13 |
| 4 | 733 | 1838 | 867.37 | 2187.52 | 18 | 20 | 29 |
| 5 | 564 | 1862 | 672.274 | 2130.69 | 14 | 19 | 23 |
| 6 | 621 | 1797 | 722.073 | 2067.21 | 13 | 18 | 23 |
| 7 | 385 | 1107 | 502.617 | 1481.64 | 9 | 12 | 17 |
| 8 | 324 | 910 | 471.475 | 1450.53 | 10 | 19 | 17 |
| 9 | 166 | 283 | 319.321 | 832.968 | 7 | 10 | 10 |


876_z76c1-2: distributions, pre-validation, px
Lacuna area (n = 9)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 54352)
count
width (px)